CATALYTIC OXIDATION OF SULFUR DIOXIDE

EFFECT OF DIFFUSION

R. W. OLSON, R. W. SCHULER, and J. M. SMITH

Purdue University, Lafayette, Indiana

The importance of diffusion in the catalytic oxidation of sulfur dioxide was determined experimentally by measuring the rate of oxidation in a flow reactor operated at various mass velocities. The catalyst consisted of platinum, coated on ⅛ in. cylindrical pellets. The reactor I.D. was 1.5 in. and the catalyst bed depth low in order to approach differential reaction conditions. The mass velocity was varied from 147 to 514 lb./(hr.)(sq.ft.) and the temperature from 350 to 480° C.

It was found that the use of available data for the diffusion rates to the catalyst surface gave a satisfactory method of accounting for the effect of diffusion. It was found further that at high temperatures, low mass velocities, and low conversions, the pressure drop between the main gas and the catalyst surface for sulfur dioxide was as much as 25% of the partial pressure of sulfur dioxide in the gas. At low temperatures and high mass velocities the pressure drop was negligible.

THE design of equipment for gaseous reactions in which a solid catalyst is used may require consideration of the physical process of diffusion as well as the adsorption, reaction, and desorption steps at the catalyst surface. Diffusion of reactants and products between the main gas stream and the catalyst is likely to be important at high temperatures, where the surface rate of reaction is high, and at low gas mass velocities, where the mass-transfer coefficients are low. The mass-transfer coefficients are primarily dependent upon the gas mass velocity and nearly independent of temperature. On the other hand, the adsorption, reaction, and desorption steps are independent of mass velocity and vary with temperature. Because of this situation the relative importance of the mass transfer and surface processes in establishing the over-all rate of reaction can be evaluated by operating a reactor under varying conditions of mass velocity and temperature.

The purpose of this investigation was to determine the importance of diffusion by measuring reaction rates at different temperatures and mass velocities for the oxidation of sulfur dioxide with a platinum catalyst. In analyzing the experimental results, the main objective was to compare the theoretical methods for taking into account the effect of diffusion on the rate with the observed data. Apparently the only other experimental information on this subject is that of Hurt (*7*) who has introduced the concepts of the H.R.U. (height of reaction unit) and pseudo-first-order reaction mechanism and applied them to the oxidation of sulfur dioxide.

Reaction rates were also measured as a function of one composition variable, the extent of conversion of the sulfur dioxide. These data were used then to test the proposals of Uyehara and Watson (*9*) and Hurt (*7*) for the mechanism controlling the rate of reaction at the catalyst surface. However, since the partial pressures of reactants and products were not varied independently and since diffusion effects always had to be accounted for, the study of reaction mechanisms was only a secondary objective of this investigation.

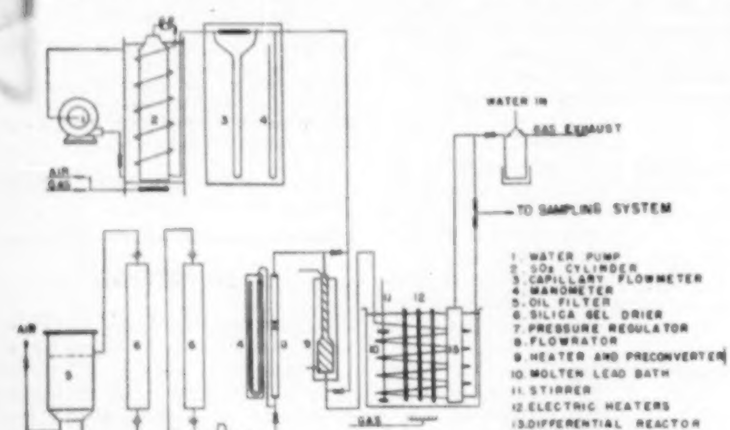


Fig. 1. Schematic diagram of apparatus.

Scope of Work

Reaction rates were measured in a 1.5-in. I.D., differential reactor packed with ⅛ in. × ⅛ in. cylindrical, alumina pellets coated with platinum. The range of operating conditions was as follows:

Temperature: 350-480° C.
Pressure: 790 mm. Hg.
Gas mass velocity: 147-514 lb./(hr.)(sq.ft. of empty reactor area)
Conversion: 4-70%

Apparatus and Procedure. The apparatus and procedure used in this investigation were similar to those discussed by Hall and Smith (*3*) and are as follows:

1. Pretreatment and metering of reactants
2. Preconversion of part of the sulfur dioxide in the preconverter
3. Measurement of the reaction rate in the differential reactor
4. Analysis of reactant and product gases

A schematic diagram of the apparatus is shown in Figure 1. Air was passed through a fiber-glass filter to remove entrained oil and then dried in two silica gel driers operated in series. The dried air was metered with a precision rotameter, 30 in. in length. The air was preheated approximately to reaction temperature in an electric heater and then passed through a preliminary converter. Sulfur dioxide was metered with a glass capillary flowmeter and also passed to the preconverter. The conversion to sulfur trioxide obtained in the preconverter permitted the investigation of rates in the presence of sulfur trioxide. Arrangements were available for admitting sulfur dioxide either before or after the preconverter, the amount admitted before being governed by the percentage of preconversion desired. The preconverter consisted of a stainless steel tube, 4 in. in length and 1.5 in. in diameter, fitted with retaining screens to hold the catalyst in place. A chromel-alumel thermocouple placed in the catalyst bed permitted maintaining a constant temperature sufficiently high to ensure approximately complete conversion.

The reactants passed through a 1-in. stainless steel coil packed with ½-in. Raschig rings and immersed in a constant-temperature, lead bath. The stainless steel differential reactor was 1.5-in. I.D. and packed with catalyst to a depth of only ⅜ in. in order to limit the conversion to low values and so obtain approximately differential rates. Details of the reactor are shown in Figure 2. Three chromel-alumel thermocouples, with leads contained in ⅛-in. stainless-steel tubes open at both ends, were used to measure the center and edge catalyst temperatures and the exit gas temperature. Thermocouple junctions used to measure catalyst temperatures were imbedded in the ⅛-in. pellets.

After leaving the differential reactor, the product gases passed through a stainless steel valve, used to control the system pressure at an average value of 790 mm., and finally through a cooler.

Samples of the reactant and product gases were withdrawn through stainless steel pipes on either side of the differential reactor and analyzed by the volumetric oxidation method described by Hall and Smith (3).

The catalyst consisted of 0.2 wt. % platinum coated on ⅛ by ⅛ in. cylindrical alumina pellets. The catalyst bed normally used contained 10.5 g. and was three pellets deep. Some runs were made with a mixture of 5.0 g. of catalyst and 5.5 g. of plain alumina pellets.

It was found that the activity of the catalyst decreased during the first day's operations (approximately 7 hr.) and reached a constant activity on the following day. It was found further that the reduction in activity was determined by the time measured from the initial contact with sulfur dioxide rather than the total on-stream time. A plot of reaction rate vs. on-stream is shown in Figure 3. Moisture had a pronounced poisoning effect on the catalyst, and, as a result, careful regeneration of the drying agent was required. To minimize the moisture problems, refrigeration or research grade sulfur dioxide was used in all the runs.

Experimental Results. Differential reaction rate data at the four mass velocities (based upon the cross-sectional area of the empty reactor) are shown in Tables 1-4. These data were obtained using but one initial concentration of reactants entering the preconverter (6.42 mole % SO_2 and the remainder air). However, the effect of restricted composition changes on the rate were determined by runs made at 0—70% preconversion obtained in the preconverter

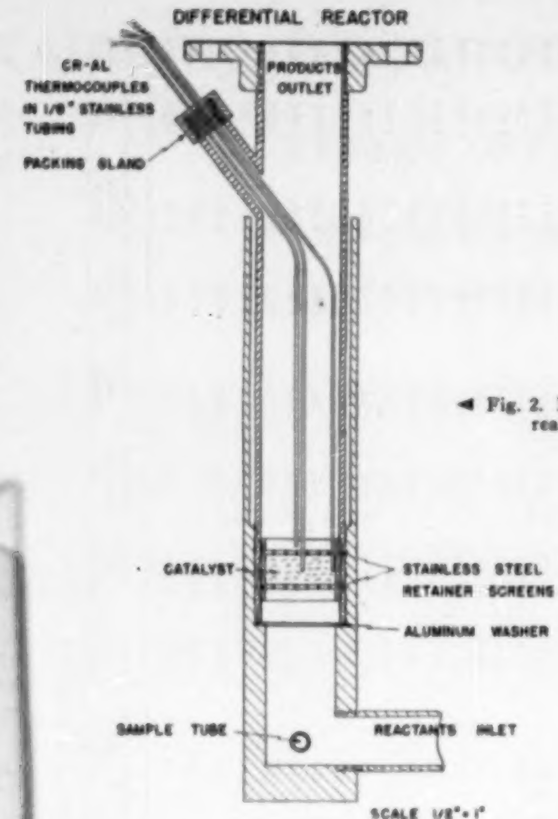
TABLE 1—DIFFERENTIAL REACTION RATE DATA

MASS VELOCITY = 147 lbs./(hr.)(sq.ft.)

Run No.	Lead Bath Temperature °C	Center Catalyst Temperature °C	Edge Catalyst Temperature °C	Center Gas Temperature °C	Pre-Converter Temperature °C	Gram Moles of SO_2/Min.	Gram Moles of Air/Min.	Mass Velocity lbs./(hr.)(sq.ft.)	% SO_2 in Reactants	% Conversion in Preconverter	% Total Conversion	% Conversion in Diff. Reactor	Weight of Catalyst (grams)	Mean Bed Temperature °C	Reaction Rate g.moles/(hr.)(g.catalyst)
B-6	344.1	354.8	351.2	350.2	473.	0.0281	0.410	147.	6.12	0.0	8.32	8.32	10.5	353.	0.0133
B-7	387.8	388.4	381.2	379.9	477.	[illegible]	0.409	147.	6.147	0.0	12.72	12.72	10.5	385.	0.0205
B-8	384.0	421.9	408.0	400.3	482.	0.0285	0.411	148.	6.22	0.0	18.84	18.84	10.5	400.	0.0307
B-9	404.5	445.0	431.1	427.6	474.	0.0284	0.411	148.	6.21	0.0	28.49	28.49	10.5	431.	0.0463
B-10	435.1	495.0	475.0	465.8	473.	0.0285	0.411	148.	6.16	0.0	39.84	39.84	10.5	476.	0.0650
B-12	348.0	363.0	359.2	358.0	515.	0.0283	0.409	147.	6.46	4.63	10.80	5.17	10.5	356.	0.010
B-13	376.0	397.0	389.7	388.1	517.	0.028	0.408	146.	6.45	4.55	18.05	13.50	10.5	387.	0.0218
B-14	391.0	420.8	410.8	408.2	516.	0.0282	0.408	147.	6.45	4.48	24.32	19.84	10.5	409.	0.0321
B-15	366.0	371.5	369.6	368.8	508.	0.0283	0.410	147.	6.45	24.2	28.15	3.95	10.5	364.	0.0064
B-16	386.2	397.5	393.9	392.3	507.	0.0283	0.409	147.	6.45	24.2[1]	31.7	7.48	10.5	389.	0.0121
B-17	416.0	438.9	431.5	428.5	507.	0.0285	0.409	147.	6.50	24.2	40.15	15.90	10.5	429.	0.0259
B-18	447.1	479.1	470.2	464.3	507.	0.0286	0.410	147.	6.49	24.2[1]	48.45	24.23	10.5	468.	0.0396
B-19	387.0	410.8	402.9	401.0	500.	0.0283	0.410	147.	6.44	11.3	26.39	15.09	10.5	400.	0.0244
B-20	407.9	440.9	430.0	426.2	500.	0.028	0.410	147.	6.43	10.9[1]	33.75	22.83	10.5	429.	0.0369
B-21	438.0	480.9	468.9	461.9	499.	0.0282	0.410	147.	6.45	10.5	42.37	31.65	10.5	468.	0.0511
B-22	389.0	398.4	395.1	394.0	500.	0.0284	0.411	148.	6.47	35.4	42.52	6.74	10.5	390.	0.011
B-23	417.7	436.0	430.0	427.6	500.	0.0285	0.411	148.	6.48	35.4[1]	48.2	12.81	10.5	426.	0.0209
B-24	448.6	473.0	466.0	461.0	500.	0.0285	0.412	148.	6.47	35.0	53.1	18.1	10.5	463.	0.0295
B-25	407.4	410.3	409.3	409.5	518.	0.0283	0.411	147.	6.45	66.4	68.3	1.9	10.5	410.	0.0031
B-26	437.4	443.1	441.5	441.1	517.	0.0283	0.411	147.	6.45	66.3[1]	70.43	4.13	10.5	442.	0.0067
B-27	475.5	484.4	482.0	480.6	517.	0.0284	0.410	147.	6.48	66.2	73.1	6.9	10.5	483.	0.0112
B-28B	373.8	389.2	382.1	——	436.	0.0284	0.409	147.	6.43	0.0	8.98	8.98	5.1	386.	0.0298
B-29B	389.3	410.8	400.8	——	453.	0.0281	0.408	146.	6.42	0.0	13.06	13.06	5.1	406.	0.0433
B-30B	433.9	444.3	430.5	——	453.	0.0279	0.408	146.	6.38	0.0	18.0	17.97	5.1	438.	0.0591
B-31B	443.2	482.6	465.8	——	453.	0.0279	0.408	146.	6.38	0.0	24.6	24.6	5.1	475.	0.0809

[1] Average preconversion from 2 runs.

Average Mass Velocity = 147.



◄ **Fig. 2. Differential reactor.**

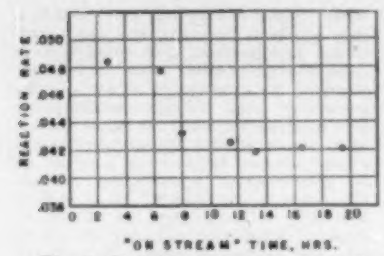


Fig. 3. Reaction rate as function of "on stream" time. Temperature 411° C. Mass velocity—245 lb./(hr.) (sq.ft.)

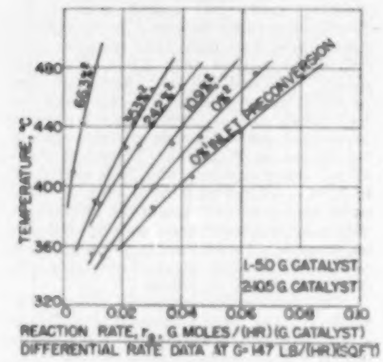


Fig. 4.

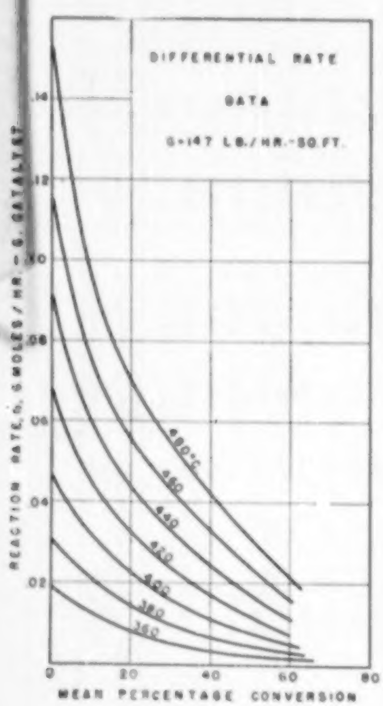


Fig. 5.

upstream from the differential reactor. Thus the composition changes can be expressed in terms of one variable, the extent of conversion.

Because of the highly exothermic reaction, it was impossible to maintain the catalyst bed at an exactly uniform temperature. The thermocouples at the edge and center indicated temperature differences of several degrees, especially at high temperatures and low preconversions. In order to represent the data accurately as a function of temperature, the following method was employed to calculate the best average temperature of the bed: a parabola was chosen to represent the radial temperature curve and then fitted to the known temperatures at the center and edge of the bed. The area-average temperature was evaluated by integration. The parabola was chosen because it was found (*3*) to fit experimental curves rather well.

The errors in the rate measurements varied with the temperature and conversion. At a low temperature, the small conversions tended to decrease the accuracy of the analytical method but increase the accuracy of the average temperature calculation. At low mass velocities, the low space velocities resulted in relatively high conversions and radial temperature gradients, which increased the deviation from differential conditions in the catalyst bed. The lower weight of active catalyst (5.0 g.) was used under these conditions in order to minimize this problem.

Using the original data in Tables 1-4, plots were made of the reaction rate versus temperature with parameters of constant

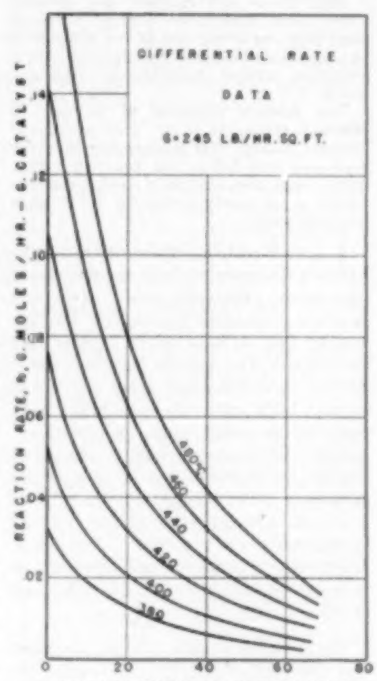


Fig. 6.

incoming conversion; that is, constant conversion entering the differential reactor. A typical plot is shown in Figure 4 for a mass velocity of 147 lb./(hr.)(sq.ft.). Values from these curves were then cross-plotted as reaction rate versus mean per cent conversion, with lines of constant temperature. Results are shown in Figures 5-8. Use of mean conversion (the average of inlet and exit conversions from the differential reactor) gives a closer approach to truly differential reaction rates than the original incoming preconversion. Arithmetic averaging was used to evaluate the mean conversion, since the exit values were always less than twice the inlet conversions (except for those runs made with 0% SO_3 entering the differential reactor).

Data in Figures 5-8 show that the presence of small amounts of sulfur trioxide produces a pronounced retardation of the reaction. This is in qualitative agreement with the earlier results of Bodenstein and Fink (*2*) and Lewis and Ries (*8*).

Figure 9 is a cross-plot of the data in Figures 5-8 showing the effect of mass velocity on the rate at a constant conversion of 10%. As expected, the curves tend to approach a constant rate as the mass velocity increases, indicating the decreasing effect of diffusion resistances at high velocities. This figure also shows that the relative importance of mass velocity becomes less as the temperature goes down.

Methods of Evaluating Effect of Diffusion. There are several methods by which rate data at different mass veloci-

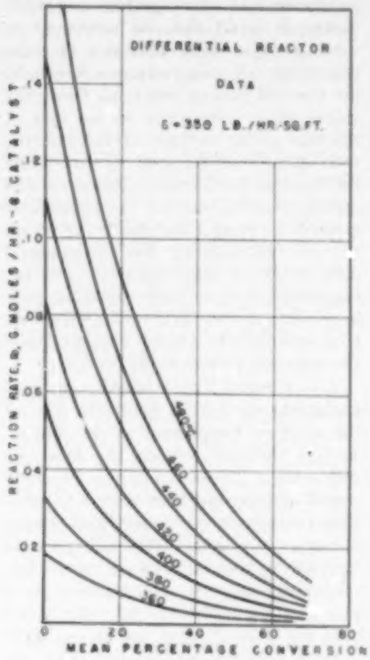


Fig. 7.

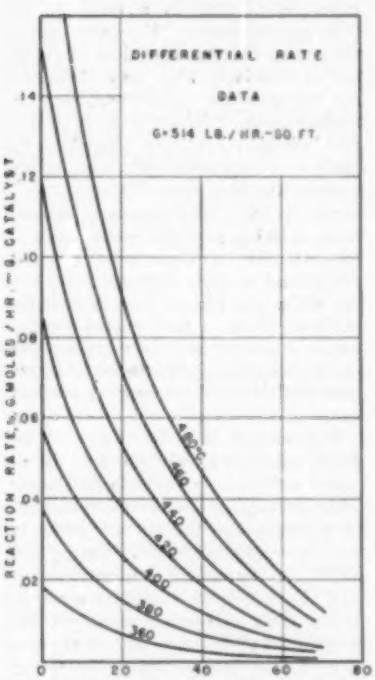


Fig. 8.

TABLE 2

DIFFERENTIAL REACTION RATE DATA

MASS VELOCITY = 244 lbs./(hr.)(sq.ft.)

Run No.	Lead Bath Temperature °C	Center Catalyst Temperature °C	Edge Catalyst Temperature °C	Center Gas Temperature °C	Pre-Converter Temperature °C	Gram Moles of SO_2/Min.	Gram Moles of Air/Min.	Mass Velocity lbs./(hr.)(sq.ft.)	% SO_2 in Reactants	% Conversion in Preconverter	% Total Conversion	% Conversion in Diff. Reactor	Weight of Catalyst (Grams)	Mean Bed Temperature	Reaction Rate g.moles/(hr.)(g.catalyst)
C-71	389.8	417.2	407.9	409.4	504.	0.0473	0.682	246.	6.36	0.0	15.7	15.7	10.5	413.	0.0424
C-72	345.5	353.7	350.1	350.6	481.	0.0471	0.683	245.	6.45	0.0	4.86	4.86	10.5	352.	0.0130
C-73	360.5	381.9	377.3	378.0	471.	0.0470	0.683	245.	6.45	0.0	8.48	8.48	10.4	380.	0.0232
C-74	433.5	452.5	439.4	441.3	485.	0.0472	0.682	245.	6.47	0.0	25.3	25.3	10.5	447.	0.0681
C-75	348.0	348.4	348.2	347.9	516.	0.0471	0.682	245.	6.46	30.4	30.8	0.43	10.5	348.	0.0011
C-76	379.3	381.7	380.6	380.4	514.	0.0470	0.682	245.	6.45	30.1	32.1	2.05	10.5	381.	0.0059
C-78	375.9	390.3	387.4	388.0	482.	0.0471	0.680	245.	6.46	0.0	10.2	10.2	10.5	390.	0.0275
C-79	398.9	406.7	404.3	404.5	509.	0.0470	0.683	245.	6.44	29.2	33.9	4.65	10.5	406.	0.0125
C-80	427.1	442.8	437.6	438.0	505.	0.0470	0.683	245.	6.45	29.2	39.1	9.85	10.5	441.	0.0264
C-81	395.1	398.3	397.0	396.7	512.	0.0469	0.682	245.	6.44	13.9	16.7	2.79	10.5	398.	0.0075
C-82	379.5	387.7	384.5	384.5	512.	0.0471	0.682	245.	6.47	13.9	19.8	5.15	10.5	386.	0.0138
C-83	412.0	430.7	425.0	425.5	512.	0.0470	0.682	247.	6.45	13.9[1]	26.3	12.4	10.5	429.	0.0331
C-84	436.6	467.1	456.5	456.8	512.	0.0472	0.682	245.	6.47	13.9[1]	33.5	19.6	10.5	462.	0.0525
C-85	448.7	472.2	466.2	463.9	496.	0.0470	0.681	245.	6.45	28.9[1]	43.3	14.4	10.5	469.	0.0388
C-86	398.9	406.2	403.8	403.7	525.	0.0470	0.682	245.	6.45	42.4	46.6	4.2	10.5	405.	0.0112
C-87	428.1	440.5	436.3	436.2	525.	0.0470	0.680	244.	6.45	42.4	50.2	7.78	10.5	439.	0.0209
C-88	446.4	461.3	455.9	455.9	526.	0.0471	0.679	244.	6.48	42.1	52.1	10.6	10.5	459.	0.0280
C-89	400.3	403.1	401.4	401.4	529.	0.0471	0.684	246.	6.45	61.9	63.9	2.0	10.5	402.	0.0054
C-90	437.6	443.5	440.8	440.7	530.	0.0471	0.684	246.	6.45	61.9[1]	66.2	4.3	10.5	442.	0.0116
C-91	458.6	466.0	463.0	462.6	529.	0.0471	0.683	246.	6.45	62.9	67.9	5.0	10.5	465.	0.0134
C-1258	371.1	380.5	376.7	——	493.	0.0475	0.682	245.	6.38	0.0	4.24	4.24	5.10	379.	0.0237
C-1268	400.9	418.8	411.3	——	503.	0.0475	0.682	245.	6.38	0.0	10.2	10.2	5.10	416.	0.0572
C-1270	433.7	462.9	451.9	——	499.	0.0475	0.681	245.	6.39	0.0	16.8	16.8	5.10	458.	0.0940

1 Preconversion of previous run.

Average Mass Velocity = 245.

TABLE 3.—DIFFERENTIAL REACTION RATE DATA

MASS VELOCITY = 350 lbs./(hr.)(sq.ft.)

Run No.	Lead Bath Temperature °C	Center Catalyst Temperature °C	Edge Catalyst Temperature °C	Center Gas Temperature °C	Pre-Converter Temperature °C	Gram Moles of SO_2/Min.	Gram Moles of Air/Min.	Mass Velocity lbs./(hr.)(sq.ft.)	% SO_2 in Reactants	% Conversion in Preconverter	% Total Conversion	% Conversion in Diff. Reactor	Weight of Catalyst (grams)	Mean Bed Temperature °C	Reaction Rate g.moles/(hr.)(g.catalyst)
A-7	344.4	349.8	349.0	348.2	489.	0.0677	0.972	350.	6.39	0.0	2.82	2.82	10.5	349.	0.0109
A-8	375.3	388.8	386.9	384.9	525.	0.0676	0.973	350.	6.40	0.0	8.25	8.25	10.5	388.	0.0319
A-9	401.8	427.2	423.0	419.5	497.	0.0677	0.972	350.	6.39	0.0	15.8	15.85	10.5	425.	0.0615
A-10	425.1	463.2	457.2	451.5	499.	0.0677	0.973	350.	6.395	0.0	24.2	24.22	10.5	461.	0.0938
A-11	352.9	353.3	353.6	353.0	508.	0.0678	0.975	351.	6.37	17.6	19.0	1.36	10.5	353.	0.0053
A-12	389.5	397.0	396.1	394.9	509.	0.0676	0.972	350.	6.36	16.7[1]	23.0	5.37	10.5	396.	0.0241
A-13	421.7	438.9	436.7	434.0	507.	0.0676	0.974	350.	6.36	16.7	29.8	13.02	10.5	438.	0.0504
A-14	388.7	392.1	392.5	392.7	486.	0.0680	0.972	350.	6.43	31.0	34.0	2.97	10.5	392.	0.0115
A-15	421.2	429.4	428.7	430.8	490.	0.0680	0.973	350.	6.40	31.2[2]	36.8	5.6	10.5	429.	0.0218
A-16	465.5	483.3	479.5	483.5	498.	0.0679	0.972	350.	6.40	31.4	43.4	12.0	10.5	482.	0.0465
A-17	390.5	392.0	391.8	392.1	491.	0.0680	0.978	352.	6.36	50.4	51.7	1.35	10.5	391.	0.0052
A-18	439.9	445.2	444.2	445.3	490.	0.0679	0.978	352.	6.36	50.5[3]	54.6	4.1	10.5	445.	0.0159
A-19	479.0	488.8	486.5	488.2	491.	0.0679	0.976	351.	6.37	50.6	57.3	6.7	10.5	488.	0.026
A-21	390.1	408.7	401.9	404.8	405.	0.0679	0.974	351.	6.37	0.0	10.1	10.1	10.5	406.	0.0393
A-22A	382.3	393.8	389.5	——	491.	0.0679	0.970	349.	6.36	0.0	5.50	5.5	5.1	392.	0.044
A-23B	408.4	428.4	420.0	——	484.	0.0682	0.969	349.	6.385	0.0	9.48	9.48	5.1	425.	0.0761
A-24B	423.7	447.3	438.4	——	486.	0.0679	0.969	349	6.34	0.0	11.5	11.5	5.1	443.	0.092
A-25B	444.6	473.5	462.4	——	486.	0.0679	0.969	349.	6.35	0.0	14.7	14.7	5.1	467.	0.117

Average Mass Velocity = 350

1 % Preconversion from run A-13.
2 % Preconversion is average of runs 14 and 16.
3 % Preconversion is average of runs 17 and 19.

ties can be analyzed to evaluate quantitatively the importance of diffusion. One approach would be to extrapolate r vs. G curves, such as those shown in Figure 9, in order to evaluate the rate r_∞ at infinite mass velocity. This rate would correspond to the case of no diffusion resistance and represents a maximum value. It could then be compared to rates at finite mass velocities to show how much diffusion resistance decreases the over-all rate of reaction. Since the values of r_∞ correspond to the rate of reaction at the surface of the catalyst, such data could be used to investigate the reaction mechanism. However, this method may be subject to considerable error in its application due to the necessity of extrapolating the experimental data to infinite mass velocity. The extrapolation can be made more accurate by plotting $1/r$ vs. $1/G$ rather than r vs. G as in Figure 9. On the reciprocal plot the intercept gives a value for $1/r_\infty$.

An approach that is similar involves evaluating the partial pressures (p_i) of the reaction components at the catalyst surface by extrapolating p_g data. In this scheme graphs similar to Figure 9 would be prepared with curves for different values of p_g. From these graphs corresponding values of G and p_g could be read at a fixed rate of reaction r. Extrapolation of this information on a plot of p_g' vs. $1/G$ (to $1/G = 0$) would give the desired p_i at the chosen rate.

Both previous procedures have the advantage of not necessitating auxiliary experimental data in addition to the rate measurements. However, unless the rate of reaction is studied at a number of relatively high mass velocities, the extrapolation requirement may introduce errors. When data at only a few values of G are available, the method of evaluating diffusional effects proposed by Hougen and Watson (5) is more useful. This approach depends upon utilizing available correlations of mass-transfer coefficients to calculate the partial-pressure differences between the main gas stream and the catalyst surface. Once the partial-pressure values at the surface are evaluated they can be used along with the experimental rate data to study the reaction mechanism.

The method used by Hurt (7) depends upon the assumption that the reaction mechanism is pseudo first order. Then separation of the resistances due to diffusion and the surface processes can be accomplished by using the H.T.U. to represent the former and the H.C.U. (height of a catalyst unit) the latter. This approach is similar to that proposed by Hougen and Watson in that it also depends upon the use of additional data to evaluate the diffusional resistances, that is, correlations of mass-transfer coefficients or H.T.U. values. The scheme proposed by Hurt is easier to apply but cannot be used to investigate the nature of the reaction mechanism at the catalyst surface.

Rate data obtained in this study have been analyzed by Hougen and Watson's and Hurt's methods. The extrapolation procedure has not been used because only four mass velocities were studied, and the maximum was only 514 lb./(hr.)(sq.ft.).

Film Pressure Drop ($p_g - p_i$) Due to Diffusion. In evaluating the partial pressure of reactants (SO_2 and O_2) and product (SO_3) at the catalyst surface, the mass-transfer correlations of Hougen and Wilkie (*6*) were employed. At modified Reynolds numbers below 350, their results can be written in terms of a mass-transfer coefficient k_g for SO_2 as follows:

$$(k_g)_{SO_2} = \frac{1.82\, G\left(\frac{d_pG}{\mu}\right)^{-0.51}}{M_m p_f}\left(\frac{\mu}{\rho D_{SO_2}}\right)^{-2/3} \qquad (1)$$

where D_{SO_2} represents the mean diffusivity of SO_2 in the diffusing mixture of SO_2, O_2, SO_3 and N_2.

The rate of mass transfer across the film is equal to the rate of reaction and may be expressed by the equation,

$$r = (k_g)_{SO_2} a (p_G - p_i) \qquad (2)$$

If r is the rate per unit mass of catalyst, then a becomes the catalyst particle area per unit mass. For the 1/8 in. × 1/8 in. cylindrical pellets used in this work the pellet density was 112.8 lb./(cu.ft.) and a = 5.12 sq.ft./lb.

Combination of Equations (1) and (2) gives the required expression for the pressure drop across the film.

$$(p_g - p_i)_{SO_2} = \frac{rM_m p_f}{9.32\, G}\left(\frac{d_pG}{\mu}\right)^{0.51}\left(\frac{\mu}{\rho D_{SO_2}}\right)^{2/3} \qquad (3)$$

This equation was employed to evaluate p_i values for SO_2 from the experimental rate and mass velocity. The pressure film factor p_f (analogous to the mean partial pressure of inert gas in the case of a single diffusing component in a stagnant gas) was evaluated according to the procedure proposed by Wilkie (*10*). Because of the large partial pressure of nitrogen in the reaction mixture, the p_f values did not change significantly across the film. This eliminated the necessity for trial and error procedures and simplified the calculations considerably.

The diffusivity of sulfur dioxide in the diffusing reaction mixture was also evaluated according to the procedure outlined by Wilkie (*10*). Calculations were simplified because the small composition changes of the whole gas mixture with conversion did not affect the Schmidt group.

Equations similar to (3) were used to determine ($p_g - p_i$) for O_2 and for SO_3. Results are summarized in Table 5, which shows $p_g - p_i$ and p_i for various mass velocities, temperatures, and conversion.

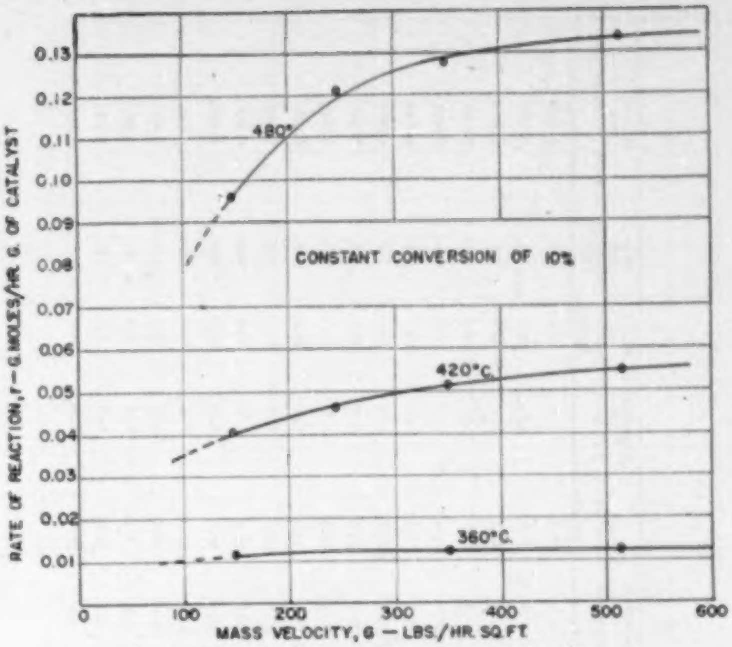


Fig. 9. Effect of mass velocity upon rate of reaction.

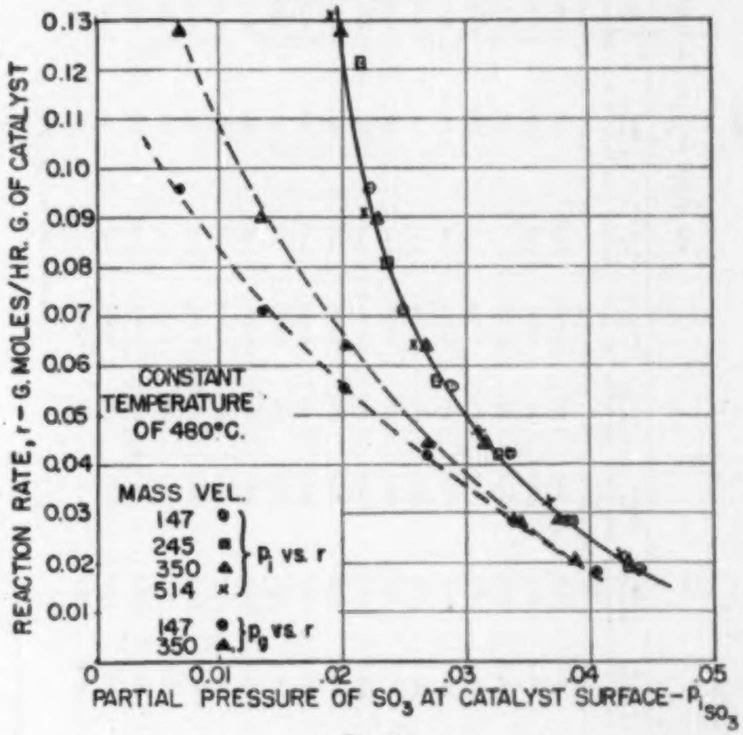


Fig. 10.

TABLE 4

DIFFERENTIAL REACTION RATE DATA

MASS VELOCITY = 514 lbs./(hr.)(sq.ft.)

Run No.	Lead Bath Temperature °C	Center Catalyst Temperature °C	Edge Catalyst Temperature °C	Center Gas Temperature °C	Pre-Converter Temperature °C	Gram Moles of SO_2/Min.	Gram Moles of Air/Min.	Mass Velocity lbs./(Hr.)(Sq.ft.)	% SO_2 in Reactants	% Conversion in Preconverter	% Total Conversion	% Conversion in Diff.Reactor	Weight of Catalyst (grams)	Mean Bed Temperature °C	Reaction Rate g.moles/(hr.)(g.catalyst)
D-55	348.9	354.8	352.3	353.2	364.	0.0980	1.45	518.	6.39	0.0	2.34	2.34	10.5	354.	0.0139
D-56	378.0	392.4	387.9	390.0	382.	0.0982	1.45	520.	6.38	0.0	6.75	6.75	10.5	390.	0.0378
D-57	404.0	430.0	422.3	424.2	422.	0.0989	1.45	521.	6.32	0.0	12.1	12.1	10.5	427.	0.0677
D-58	438.1	480.1	468.0	471.5	475.	0.0982	1.46	522.	6.32	0.0	21.2	21.2	10.5	475.	0.119
D-59	369.8	371.3	370.8	370.8	466.	0.0984	1.45	521.	6.35	20.9	22.6	0.64	10.5	371.	0.0095
D-60	402.8	410.0	406.7	408.2	466.	0.0984	1.45	522.	6.34	20.9[1]	26.3	4.28	10.5	409.	0.0241
D-61	402.8	410.4	407.0	408.3	476.	0.0982	1.42	511.	6.45	20.9[1]	25.5	4.63	10.5	409.	0.0260
D-62	432.1	448.0	442.5	444.6	477.	0.0982	1.42	512	6.44	20.9[1]	29.3	8.38	10.5	446.	0.047
D-63	458.0	480.9	472.5	475.5	477.	0.0982	1.42	511.	6.46	21.5	33.3	11.80	10.5	477.	0.0661
D-64	388.1	391.0	389.8	390.0	491.	0.0978	1.43	513.	6.41	37.6	39.4	1.80	10.5	390.	0.010
D-65	419.1	427.0	423.7	424.8	490.	0.0977	1.43	513.	6.40	37.3	41.3	4.05	10.5	426.	0.0226
D-66	402.0	405.1	404.1	405.0	496.	0.0976	1.40	511.	6.49	54.5	55.4	0.90	10.5	405.	0.005
D-67	461.4	469.5	466.7	468.2	497.	0.0976	1.40	511.	6.49	54.5[1]	58.1	3.58	10.5	468.	0.020
D-68	387.6	392.6	390.8	392.2	500.	0.0984	1.43	514.	6.43	33.8	36.2	2.41	10.5	392.	0.0136
D-69	424.8	436.0	432.3	434.4	499.	0.0981	1.43	514	6.42	34.8[1]	38.8	4.06	10.5	434.	0.0231
D-70B	391.8	403.9	400.0	——	455.	0.0981	1.42	510.	6.47	0.0	5.46	5.46	5.1	402.	0.063
D-71B	424.1	443.9	436.8	——	476.	0.0982	1.42	511.	6.47	0.0	9.11	9.11	5.1	441.	0.105
D-72B	453.7	480.0	472.1	——	472.	0.0981	1.42	510.	6.47	0.0	12.9	12.91	5.1	476.	0.149

Average Mass Velocity = 514.

1 Preconversion from previous run.

The reliability of this scheme of taking into account the effect of diffusion is illustrated in Figure 10, where the reaction rate is plotted vs. the values of p_i for SO_3 calculated by the Equation for SO_3 analogous to (3). It is apparent that the results for the four different mass velocities fall approximately on a single curve, showing that Equation (3) satisfactorily eliminates the effect of diffusion. For comparison, values of p_g are also shown on the figure (as solid points) for the two mass velocities of 147 and 350 lb./(hr.)(sq.ft.). The deviation between these two p_g vs. r curves is considerable, illustrating the need for correcting for diffusion. A similar plot showing p_i for SO_2 is presented in Figure 11. As expected from the results of Figure 10, the points for different mass velocities again fall close to a single curve.

The criterion that the data for all mass velocities must fall on a single curve in Figure 10 and 11, if diffusion is to be accounted for satisfactorily, may not be immediately apparent. It is based upon the existence of a relationship between p_i values for SO_2, SO_3, and O_2 which is nearly independent of mass velocity. Such a relationship did exist in this investigation because of two things: (1) the composition of the reaction mixture was varied only by conversion, and (2) the particular values of the diffusivities of SO_2, SO_3, and O_2. For example, if Equation (3) is applied to SO_2 and SO_3 and the p_g quantities eliminated by the stoichiometry of the reaction, the following equation results:

$$p_{iSO_3} = \frac{X_o p_t - p_{iSO_2}}{1 - .5X_o} + \left(K_{SO_3} - \frac{K_{SO_2}}{1 - 0.5X_o}\right) rG^{-0.49} \quad (4)$$

where

$$K_{SO_3} = \frac{M_m p_f}{1.82a}\left(\frac{d_p}{\mu}\right)^{0.51}\left(\frac{\mu}{\rho D_{SO_3}}\right)^{2/3} \quad (5)$$

$$K_{SO_2} = \frac{M_m p_f}{1.82a}\left(\frac{d_p}{\mu}\right)^{0.51}\left(\frac{\mu}{\rho D_{SO_2}}\right)^{2/3} \quad (6)$$

The value of X_o, the mole fraction of sulfur dioxide in the gas with no conversion, was 0.0642. Since $K_{SO_3} = 1.07\ K_{SO_2}$, approximately, the value of the group in parentheses in Equation (4) was only about 0.04 K_{SO_3}. This, coupled with the fact that $K_{SO_3} rG^{-0.49}$ was quite small, made Equation (4) insensitive to changes in mass velocity. Hence, the relationship between P_{iSO_3} and P_{iSO_2} is determined by the first two terms of the equation. A similar situation existed with respect to p_{iO_2} and p_{iSO_2}.

The magnitude of the pressure drop due to diffusion resistance is illustrated in Figures 12 and 13, which were prepared from the data in Table 5. Figure 12 shows the effect of temperature and mass velocity upon the ratio of the pressure drop across the film to the pressure in the main gas stream, that is, $(p_g - p_i)/p_g$.

As pointed out by Yang and Hougen (5), plots of this type are valuable in determining under what conditions diffusion resistances are important. For example, if the required accuracy in the measurement of reaction rates is 10%, the value of $(p_g - p_i)/p_g$ should be 0.90 or greater. From Figure 12, this condition would be satisfied at a conversion of 10% provided the temperature were less than about 440° C. at $G = 514$, and less than 420° C. at $G = 147$ lb./(hr.)(sq.ft.). Figure 13 shows that at 480° C., for the same accuracy, the conversion would have to be greater than 35% at $G = 514$ and greater than 64% at $G = 147$ lb./(hr.)(sq.ft.).

H.R.U. Approach. Hurt (7) defined the H.R.U. in terms of the depth of catalyst bed h and the partial pressure of one of the reactants as follows:

$$\text{H.R.U.} = \frac{h}{-\int_{p_{g1}}^{p_{g2}} \frac{dp_g}{p_g - p^*}} \qquad (7)$$

where p^* is the equilibrium value of the partial pressure at the temperature of the reactor. This equation, applied to the differential reactor data obtained in this investigation, becomes

$$\text{H.R.U.} = \frac{h}{\ln \dfrac{p_{g1} - p^*}{p_{g2} - p^*}} \qquad (8)$$

Equation (8) was used to evaluate H.R.U. values based upon the partial pressures of sulfur dioxide. The measured conversion at the entrance and exit of the differential catalyst bed permitted the evaluation of p_{g1} and p_{g2}. Equilibrium partial pressures p^* were determined from the following equilibrium equation

$$K = e\left(\frac{22{,}200}{RT} - \frac{20.96}{R}\right) \qquad (9)$$

Results are illustrated in Table 6 which includes besides the H.R.U., values of p_{gSO_3} at various conversions, temperatures, and mass velocities.

To determine the effect of diffusion on the H.R.U., Hurt proposed a separation into two terms: one, the H.T.U., expressing the resistance to diffusion in the film surrounding the catalyst pellet; and the other, the H.C.U., expressing the resistance due to the processes occurring at the catalyst surface. To

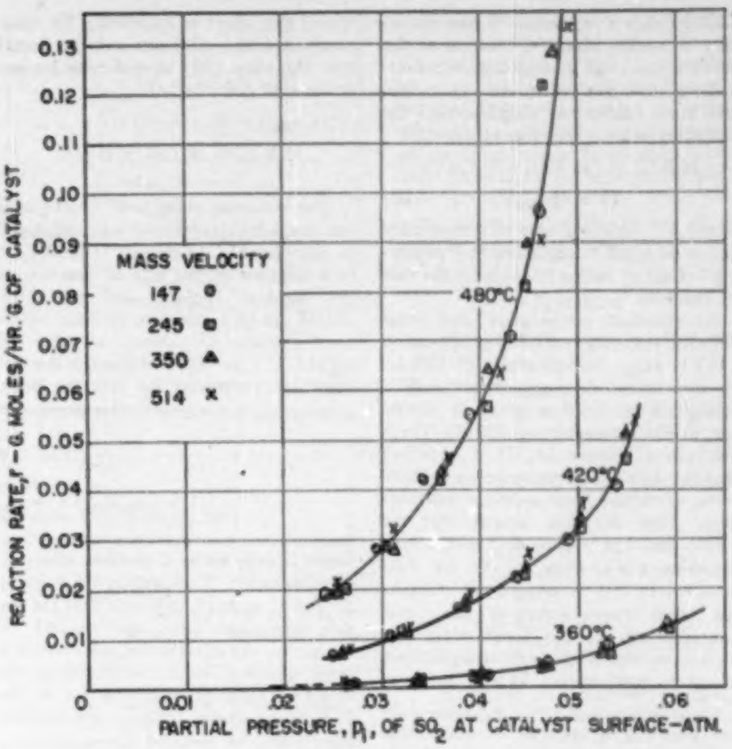


Fig. 11.

Table 5

[illegible]

Conversion Percent		G = 147 lbs./(hr.)(sq.ft.) 360° C $p_g - p_i$	360° C p_i	420° C $p_g - p_i$	420° C p_i	480° C $p_g - p_i$	480° C p_i	G = 245 lbs./(hr.)(sq.ft.) 360° C $p_g - p_i$	360° C p_i	420° C $p_g - p_i$	420° C p_i	480° C $p_g - p_i$	480° C p_i
10	SO_2	0.0018	0.0585	0.0061	0.0562	0.0141	0.0468	0.0023	0.0580	0.0054	0.0549	0.0140	0.0463
	SO_3	0.0020	0.0087	0.0066	0.0133	0.0153	[illegible]	0.0025	[illegible]	0.0059	0.0126	0.0152	[illegible]
	O_2	[illegible]	0.200	[illegible]	0.199	[illegible]	0.196	[illegible]	[illegible]	[illegible]	0.199	[illegible]	[illegible]
20	SO_2	0.0012	0.0527	0.0044	0.0495	0.0104	0.0435	0.0016	0.0523	0.0037	[illegible]	0.0093	0.0446
	SO_3	0.0013	0.0148	0.0048	0.0183	0.0113	0.0248	0.0017	0.0153	0.0040	0.0175	0.0101	0.0236
	O_2	[illegible]	0.197	0.0015	0.196	[illegible]	0.194	0.0005	0.198	[illegible]	0.197	0.0032	0.195
30	SO_2	0.0007	0.0486	0.0034	0.0439	0.0081	0.0392	0.0011	0.0462	0.0027	0.0446	0.0066	0.0407
	SO_3	0.0008	0.0210	0.0037	0.0239	0.0088	0.0291	0.0012	0.0214	0.0029	0.0232	0.0071	0.0273
	O_2	[illegible]	0.195	[illegible]	0.194	[illegible]	[illegible]	[illegible]	0.195	[illegible]	0.194	[illegible]	0.193
40	SO_2	[illegible]	[illegible]	0.0025	0.0381	[illegible]	0.0344	0.0007	[illegible]	0.0019	0.0387	0.0049	[illegible]
	SO_3	[illegible]	0.0276	0.0027	[illegible]	[illegible]	0.0339	0.0008	0.0279	0.0021	[illegible]	0.0052	[illegible]
	O_2	[illegible]	0.192	[illegible]	0.191	[illegible]	0.190	[illegible]	0.192	[illegible]	0.191	[illegible]	[illegible]
50	SO_2	0.0003	0.0336	0.0017	[illegible]	0.0041	[illegible]	0.0005	0.0334	0.0013	0.0326	0.0032	0.0307
	SO_3	0.0003	0.0343	[illegible]	0.356	0.0045	[illegible]	0.0005	0.0346	0.0014	0.0354	0.0035	[illegible]
	O_2	[illegible]	0.189	[illegible]	0.189	[illegible]	[illegible]	[illegible]	0.189	[illegible]	0.189	[illegible]	[illegible]
60	SO_2	0.0001	0.0272	0.0011	0.0262	[illegible]	0.0245	0.0003	0.0270	0.0009	[illegible]	0.0022	0.0254
	SO_3	[illegible]	[illegible]	0.0012	0.0421	0.0030	0.0440	0.0004	0.0413	0.0009	0.0419	0.0024	0.0431
	O_2	[illegible]	0.187	[illegible]	0.187	[illegible]	[illegible]	[illegible]	0.187	[illegible]	0.187	[illegible]	[illegible]

Conversion Percent		G = 350 lbs./(hr.)(sq.ft.) 360° C $p_g - p_i$	360° C p_i	420° C $p_g - p_i$	420° C p_i	480° C $p_g - p_i$	480° C p_i	G = 514 lbs./(hr.)(sq.ft.) 360° C $p_g - p_i$	360° C p_i	420° C $p_g - p_i$	420° C p_i	480° C $p_g - p_i$	480° C p_i
10	SO_2	0.0012	0.0587	0.0051	0.0548	0.0124	0.0475	0.0010	0.0591	0.0044	0.0557	0.0108	0.0493
	SO_3	0.0013	0.0080	0.0056	0.0122	0.0134	[illegible]	0.0011	0.0078	0.0048	0.0115	0.0117	0.0183
	O_2	[illegible]	0.201	[illegible]	[illegible]	[illegible]	0.197	0.0003	[illegible]	[illegible]	0.200	0.0037	0.197
20	SO_2	[illegible]	0.0527	0.0034	0.0501	[illegible]	0.0447	0.0007	0.0530	[illegible]	0.0505	0.0073	0.0464
	SO_3	[illegible]	0.0143	0.0037	0.0171	[illegible]	0.0230	0.0007	[illegible]	[illegible]	[illegible]	0.0079	0.0213
	O_2	[illegible]	0.198	[illegible]	0.197	0.0030	0.195	0.0002	0.198	0.0011	0.197	0.0025	0.196
30	SO_2	0.0005	0.0465	0.0023	0.0447	[illegible]	[illegible]	0.0004	[illegible]	[illegible]	0.0450	0.0051	0.0421
	SO_3	0.0005	[illegible]	0.0025	0.0227	[illegible]	[illegible]	0.0004	[illegible]	0.0024	[illegible]	0.0055	0.0257
	O_2	[illegible]	0.195	[illegible]	0.195	0.0022	0.193	0.0002	0.195	[illegible]	0.194	0.0017	0.193
40	SO_2	0.0003	0.0400	0.0016	0.0387	0.0043	0.0360	0.0002	[illegible]	0.0015	0.0390	0.0037	[illegible]
	SO_3	0.0003	0.0272	0.0017	[illegible]	0.0046	0.0316	0.0002	0.0273	0.0017	0.0287	[illegible]	0.0313
	O_2	[illegible]	0.193	[illegible]	0.192	0.0015	0.191	[illegible]	0.192	[illegible]	0.192	0.0013	0.191
50	SO_2	[illegible]	0.0335	0.0011	0.0326	0.0027	0.0310	[illegible]	[illegible]	0.0009	0.0329	[illegible]	0.0312
	SO_3	[illegible]	[illegible]	0.0012	0.0350	0.0029	[illegible]	[illegible]	[illegible]	0.0011	0.0350	0.0028	0.0367
	O_2	[illegible]	0.190	[illegible]	0.189	[illegible]	0.188	[illegible]	0.190	[illegible]	0.189	[illegible]	0.188
60	SO_2	—	0.0270	0.0007	0.0264	0.0018	0.0253	0.0001	0.0271	[illegible]	0.0266	0.0017	0.0255
	SO_3	—	[illegible]	0.0008	[illegible]	0.0020	0.0426	0.0001	[illegible]	[illegible]	0.0415	0.0018	0.0426
	O_2	—	0.187	[illegible]	0.187	[illegible]	0.187	—	0.187	[illegible]	0.187	[illegible]	[illegible]

achieve such a separation it was necessary to assume that the reaction at the surface was controlled by a pseudo-first-order mechanism. With this assumption and some additional simplification the following relationship was obtained:

$$\text{H.R.U.} = (\text{H.T.U.} + \text{H.C.U.})(1 + k p_{gSO_3}) \quad (10)$$

where k is function only of temperature and is designed to represent the "poisoning" effect of sulfur trioxide on the rate of reaction.

At constant temperature and mass velocity the only variable in Equation (10) is p_{SO_3}. Hence, a plot of H.R.U. vs. conversion, or p_{gSO_3}, should yield a straight line. Such a graph is shown for a mass velocity of 350 lb./(hr.)(sq.ft.) in Figure 14. It is observed that the data at all temperatures indicate some deviation from a linear relationship. This deviation means that the single constant pseudo first order, rate Equation (10), does not fit the data obtained in this investigation.† Graphs for other mass velocities are similar, although at the lower mass velocities the deviation from straight lines is not as pronounced. It may be mentioned that Hurt in examining the same reaction with a platinum catalyst found that Equation (10) agreed with his experimental information

At $p_{gSO_3} = 0$,

$$\text{H.R.U.} = \text{H.T.U.} + \text{H.C.U.} \quad (11)$$

This equation along with Hurt's data for mass-transfer rates was employed to calculate H.C.U. values. The H.C.U. is a measure of the rate of reaction at the catalyst surface, and therefore, should not be a function of mass velocity. Figure 15 shows a plot of ln(H.C.U.) vs. $1/T$. Although the results are correlated by straight lines, there is seen to be a distinct effect of mass velocity. Comparison of Figures 10 and 11 with 15, indicates that for the data presented here, the method of evaluating surface partial pressures is a more accurate means of handling the effect of diffusion than the use of the H.R.U. approach. This is not entirely unexpected in view of the assumption regarding reaction mechanism that is necessary in applying the latter scheme.

Reaction Mechanism. As stated earlier, insufficient composition variables were studied to make a thorough study of the controlling mechanism for the reaction at the catalyst surface. However, various mechanisms leading to expressions for the rate, have been proposed for this reaction (*2, 8, 9*), and the data presented in this paper can be used to check these earlier proposals. The most fundamental approach appears to be that of Uyehara and Watson (*9*) who used the data of Lewis and Ries to test various mechanisms. Their analysis indicated that the most likely rate-controlling step in the over-all process at the catalyst surface was the reaction between adsorbed sulfur dioxide and adsorbed atomic oxygen. The rate equation corresponding to this case is

$$r = \frac{E_A C T e^{-\frac{\Delta H^*}{RT}}\left(p_{SO_2} p_{O_2} - \frac{p_{SO_3}}{K}\right)}{[1 + (p_{O_2} K_{O_2})^{1/2} + p_{SO_2} K_{SO_2} + p_{SO_3} K_{SO_3} + p_{N_2} K_{N_2}]^2} \quad (12)$$

where all the partial pressures in this and the following equations apply at the catalyst surface, i.e., they are p_i values. At a fixed temperature all the adsorption coefficients are constant as well as the group

$$E_A C T e^{-\frac{\Delta H^*}{RT}} \text{ (designated as } C_T^2\text{)}$$

Also the partial pressure of nitrogen was essentially constant so that its effect may be neglected. Hence, Equation (12) may be rearranged in the following way, in which the only variables are p_{SO_2}, p_{O_2}, and p_{SO_3}:

$$\left[\frac{p_{SO_2} p_{O_2}{}^{1/2} - \frac{p_{SO_3}}{K}}{r}\right]^{1/2} = \frac{1}{C_T}[1 + p_{SO_2} K_{SO_2} + p_{O_2}{}^{1/2} K_{O_2}{}^{1/2} + p_{SO_3} K_{SO_3}] \quad (13)$$

Because the only composition variable studied in this investigation was conversion, there existed a relationship between p_{SO_2}, p_{SO_3} and p_{O_2}. Therefore, Equation (13) can be written,

$$R = B + D p_{SO_3} \quad (14)$$

where

$$R = \left[\frac{p_{SO_2} p_{O_2}{}^{1/2} - \frac{p_{SO_3}}{K}}{r}\right]^{1/2}$$

and B and D are constant at a fixed temperature. Equation (14) is linear in form, and the constants B and D can be evaluated by the method of least squares. The experimental data necessary are the rate and p_i values given in Table 5 and shown for SO_3 and SO_2 in Figures 10 and 11. Values of B and D so determined at various temperatures are shown in Table 7.

The fact that the values of B and D in Table 7 are all positive indicates that the proposed mechanism gives a satisfactory interpretation for the rate data

† It should be noted that Equation (10) was proposed by Hurt as an approximate simplification of the more rigorous expression

$$\text{H.R.U.} = \text{H.T.U.}(1 + k p_{gSO_3})^n + \text{H.C.U.}(1 + k p_{gSO_3})^n \quad (10a)$$

This equation contains two constants, k and n, which must be evaluated from experimental information. Data obtained in this investigation are correlated better by Equation (10a) than (10) as would be expected from the additional latitude permitted by the two-constant equation.

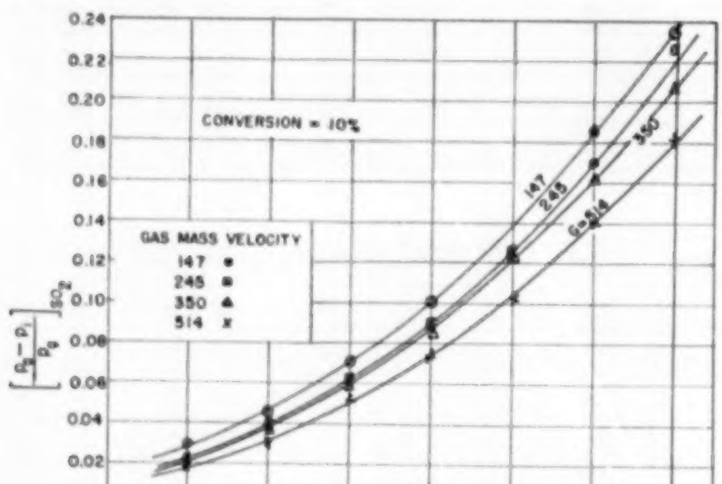


Fig. 12. Effect of temperature upon film pressure drop.

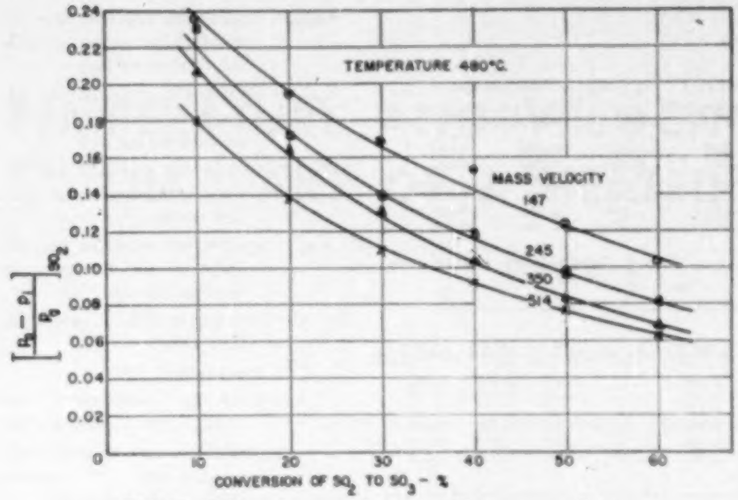


Fig. 13. Effect of conversion upon film pressure drop.

of this investigation. Positive values are a requirement because of the necessity that the adsorption coefficients in Equation (12) be greater than zero. Equation (14) is compared with the experimental rate data in Figure 16 at three temperatures. The agreement of the data with the straight line relationship is reasonably good in each case.

Five other proposed mechanisms were analyzed in the same manner. These were as follows:

1. Surface rate controlling: adsorbed atomic oxygen and sulfur dioxide in the gas phase
2. Surface rate controlling: adsorbed sulfur dioxide and molecular oxygen in the gas stream
3. Adsorption of sulfur dioxide controlling the reaction between adsorbed atomic oxygen and sulfur dioxide
4. Adsorption of oxygen controlling the reaction between adsorbed atomic oxygen and sulfur dioxide
5. Desorption of sulfur trioxide controlling the rate between adsorbed atomic oxygen and sulfur dioxide

Results of these analyses showed that both mechanisms (1) and (2) were acceptable and correlated the experimental data in tests similar to Figure 16, but (3), (4) and (5) led to negative adsorption coefficients. In summary, any of the mechanisms based upon surface reaction rate as the controlling feature was satisfactory, while those based upon adsorption or desorption as controlling were unsatisfactory.

Notation

a = catalyst area per unit mass, lb./sq.ft.

B,D = constants (at a fixed temperature) in surface reaction rate equation. Defined by Equations (13) and (14)

C_T^2 = over-all rate constant in kinetic equations

$D = C_T^2 = E_A C T e^{-\frac{\Delta H^*}{RT}}$ a constant at a fixed temperature

D = diffusivity of a single component in gas mixture, sq. ft./sec.

d_p = effective diameter of catalyst $= (1.5)^{1/3}$ (diameter of cylindrical catalyst pellet)

E_A = effectiveness factor of catalyst. Constant for a given catalyst

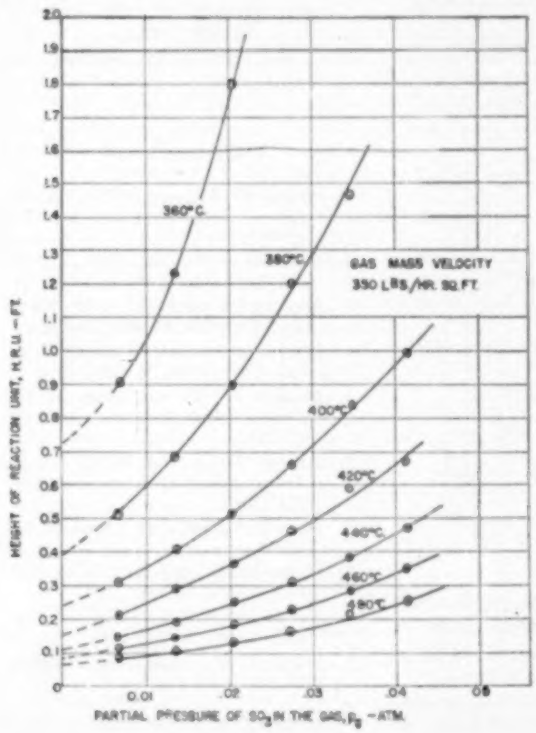


Fig. 14.

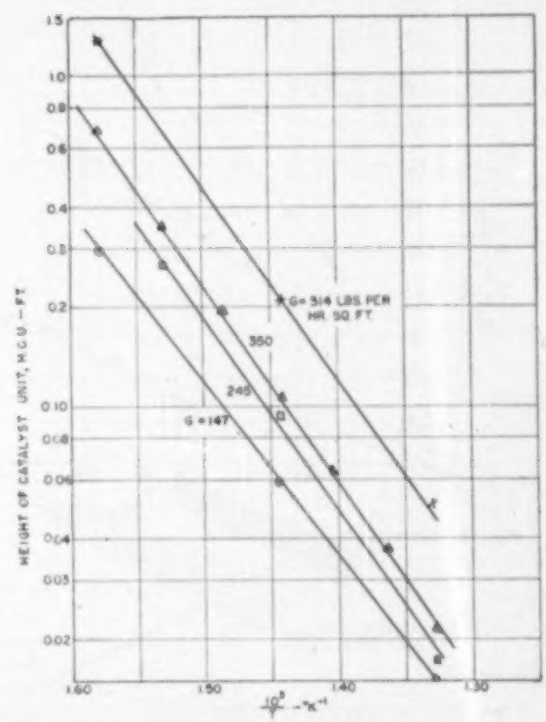


Fig. 15. H. C. U. vs. temperature.

TABLE 6

H.R.U. Values

Conversion Percent	P_{SO_3} Atm	G = 147 lbs./(hr.)(sq.ft.) HRU - ft.		G = 245 lbs./(hr.)(sq.ft.) HRU - ft.		
		360° C	420° C	360° C	420° C	480° C
10	0.0064	0.375	0.103	0.407	0.170	0.068
20	0.0131	0.497	0.127	0.526	0.219	0.081
30	0.0198	0.730	0.153	0.684	0.261	0.102
40	0.0267	0.994	0.179	0.855	0.309	0.117
50	0.0336	1.19	0.215	1.03	0.360	0.128
60	0.0406	2.22	0.282	1.17	0.386	0.139

G = 350 lbs./(hr.)(sq.ft.)

Conversion Percent	P_{SO_3} Atm.	HRU - ft.						
		360° C	380° C	400° C	420° C	440° C	460° C	480° C
10	0.0068	0.907	0.512	0.308	0.212	0.147	0.110	0.083
20	0.0131	1.22	0.681	0.407	0.292	0.192	0.142	0.107
30	0.0198	1.79	0.898	0.512	0.356	0.242	0.181	0.132
40	0.0267	2.59	1.20	0.666	0.462	0.308	0.224	0.160
50	0.0336	3.38	1.47	0.838	0.593	0.376	0.277	0.207
60	0.0406	---	---	0.99	0.667	0.470	0.344	0.244

G = 514 lbs./(hr.)(sq.ft.)

Conversion Percent	P_{SO_3} Atm.	HRU - ft.		
		360° C	420° C	480° C
10	0.0064	1.55	0.297	0.122
20	0.0131	2.25	0.391	0.153
30	0.0198	3.64	0.458	0.190
40	0.0267	6.24	0.564	0.211
50	0.0336		0.738	0.257
60	0.0406		0.978	0.317

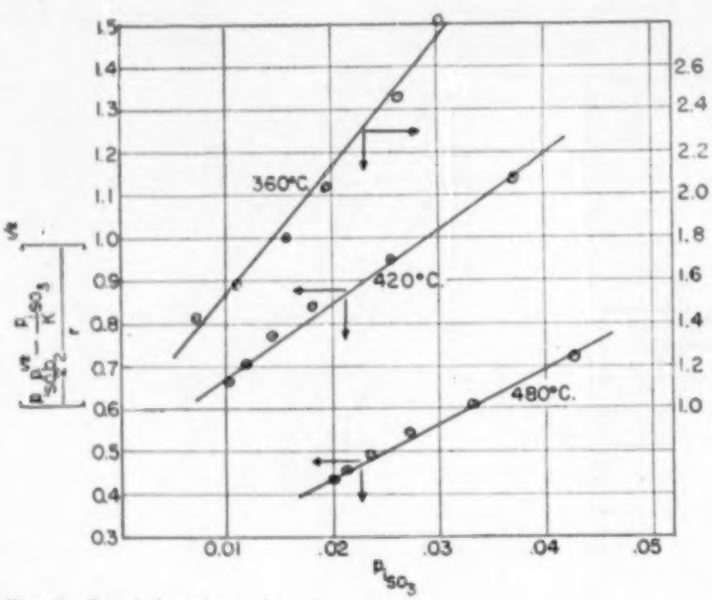


Fig. 16. Correlation of rate data. ▲

TABLE 7.—RATE CONSTANTS FOR EQUATION (14)

Temperature ° C.	B	D
360	0.906	61.6
420	0.499	17.5
480	0.176	12.9

G = gas mass velocity, lb./(hr.)(sq.ft. of empty reactor)

$\Delta H°$ = over-all standard enthalpy charge for reaction

K = equilibrium constant for over-all oxidation of sulfur dioxide

K_{SO_2} = equilibrium constant for adsorption of sulfur dioxide on catalyst surface

K_{SO_3} = equilibrium constant for adsorption of sulfur trioxide on catalyst surface

K_{O_2} = equilibrium constant for adsorption of oxygen on catalyst surface

K_{N_2} = equilibrium constant for adsorption of nitrogen on catalyst surface

n = constant in H.R.U. rate equation (10a)

T = temperature, ° K.

k = poisoning coefficient in H.R.U. rate Equation (10)

p_g = partial pressure, in main body of gas, atm.; p^* = equilibrium partial pressure

p_i = partial pressure at catalyst-gas interface, atm.

p_f = pressure film factor, atm.

p_t = total pressure, atm.

r = over-all reaction rate, gram moles of sulfur dioxide converted per hour per gram of catalyst. r_∞ rate at infinite mass velocity

X_o = mole fraction of sulfur dioxide in unconverted reaction mixture = 0.0642

μ = viscosity of gas mixture, lb./(hr.)(ft.)

ρ = density of gas mixture, lb./(cu.ft.)

Literature Cited

1. Akers, W. W., and White, R. R., *Chem. Eng. Progress*, **44**, 553 (1948).
2. Bodenstein, M., and Fink, C. G., *Z. physik Chem.*, **60**, 1 (1907).
3. Hall, R. E., and Smith, J. M., *Chem. Eng. Progress*, **44**, 459 (1948).
4. Hougen, O. A., and Watson, K. M., *Ind. Eng. Chem.*, **35**, 529 (1943).
5. Hougen, O. A., and Yang, K. H., *Chem. Eng. Progress*, **46**, 146 (1950).
6. Hougen, O. A., and Wilkie, C. R., *Trans. Am. Inst. Chem. Engrs.*, **41**, 445 (1945).
7. Hurt, D. M., *Ind. Eng. Chem.*, **35**, 522 (1943).
8. Lewis, W. K., and Ries, E. D., *Ind. Eng. Chem.*, **19**, 830 (1927).
9. Uyehara, O., and Watson, K. M., *Ind. Eng. Chem.*, **35**, 541 (1943).
10. Wilkie, C. R., *Chem. Eng. Progress*, **46**, 95 (1950).

(Presented at Forty-second Annual Meeting, Pittsburgh, Pa.)